Summary Test Maths

This covers sections of Algebra (A1 to A5) from the edexcel specification.

Topics tested (Use Maths Genie to learn if needed):

1. Simplifying Algebra (Q1 - Q6)
2. Indices - (Q7 - Q13)
3. Substitution (Q14)
4. Rearranging Harder Formulae (Q15 - Q18)
5. Expanding Triple Brackets (Q19 - Q22)

Q1.

Simplify $3x + 4x - 2x$

Q2.

Simplify $8g + 6h - 3g + h$

Q3.

(a) Simplify $a \times b \times c$

(b) Simplify $5p - 2p$

(c) Simplify $\frac{6h}{3}$

Q4.

(a) Simplify $2a \times 3b$

(b) Simplify $2p \times 2p$

(c) Simplify $\frac{7x + 5x}{4}$

Q5.

(a) Simplify $6j \times 5k$

(b) Simplify $7a - 6b + 5a + 4b$

Q6.

(a) Simplify $a^2 + a^2 + a^2$

(b) Simplify $2rs - 5rs + 4rs$

(c) Simplify $4a + 2 - 7a + a - 6$

Q7.

(a) Simplify $x^8 \times x^3$

(b) Simplify $(5y)^3$

(c) Simplify $\dfrac{w^7}{w^4}$

Q8.

(a) Simplify $2m^2 \times 5n^6$

(b) Simplify $15p^3 \div 3p^4$

Q9.

$y^2 \times y^a = y^7$

(a) Find the value of a.

$(y^4)^b = y^{12}$

(b) Find the value of b.

Q10.

Work out the value of $\dfrac{2^9 \times 2^{-2}}{2^4}$

Q11.

(a) Simplify $9p^3 \times 2p^{-2}$

Q12.

(a) Simplify $5c^2d^3 \times 2d$

(b) Write 64×4^5 as a power of 4

Q13.

Simplify $\dfrac{10p^3q^5r}{4p^3q^6}$

Q14.

$$s = \frac{v^2 - u^2}{2a}$$

$v = 7$
$u = 5$
$a = 3$

Work out the value of s.

Q15.

Make a the subject of the formula $v = u + at$

Q16.

Make v the subject of the formula $T = \frac{1}{2}mv^2$

Q17.

Make x the subject of the formula $a = \frac{x + 4}{x + 2}$

Q18.

Make x the subject of the formula $\frac{a}{b} = \frac{2x}{x + 5}$

Q19.

Expand and Simplify $(x + 2)(x + 4)(x + 1)$

Q20.

Expand and Simplify $(x + 3)(x - 1)^2$

Q21.

Expand and Simplify $(2x - 3)(x - 2)(3x - 1)$

Q22.

Show that $(5x - 4)(3x + 1)(2x - 7) = 30x^3 - 119x^2 + 41x + 28$

for all values of x.

Answer :

A1.

Simplify $3x + 4x - 2x$

$7x - 2x$

$5x$

A2.

Simplify $8g + 6h - 3g + h$

$5g + 7h$

A3.

(a) Simplify $a \times b \times c$

abc

(1)

(b) Simplify $5p - 2p$

$3p$

(1)

(c) Simplify $\frac{6h}{3}$

$2h$

(1)

A4.

(a) Simplify $2a \times 3b$

$6ab$

(1)

(b) Simplify $2p \times 2p$

$4p^2$

(1)

(c) Simplify $\frac{7x + 5x}{4}$

$\frac{12x}{4}$

$3x$

(1)

A5.

(a) Simplify $6j \times 5k$

$30jk$

(1)

(b) Simplify $7a - 6b + 5a + 4b$

$12a - 2b$

(2)

A6.

(a) Simplify $a^2 + a^2 + a^2$

$3a^2$

(1)

(b) Simplify $2rs - 5rs + 4rs$

$-3rs + 4rs$

rs

(1)

(c) Simplify $4a + 2 - 7a + a - 6$

$-2a - 4$

(2)

A7.

(a) Simplify $x^8 \times x^3$

x^{11}

(1)

(b) Simplify $(5y)^3$

$5y \times 5y \times 5y$

$125y^3$

(1)

(c) Simplify $\frac{w^7}{w^4}$

w^3

(1)

A8.

(a) Simplify $2m^2 \times 5n^6$

$10m^2n^6$

(1)

(b) Simplify $15p^3 \div 3p^4$

$5p^{-1}$

(2)

A9.

$y^2 \times y^a = y^7$

(a) Find the value of a.

5

(1)

$(y^4)^b = y^{12}$

(b) Find the value of b.

3

(1)

A10.

Work out the value of $\dfrac{2^9 \times 2^{-2}}{2^4}$

$$\frac{2^7}{2^4} = 2^3 = 8$$

$\cancel{2^3}$ 8

A11.

(a) Simplify $9p^3 \times 2p^{-2}$

18p

A12.

(a) Simplify $5c^2d^3 \times 2d$

$10c^2d^4$

(1)

(b) Write 64×4^5 as a power of 4

$$4^3 \times 4^5 = 4^8$$

4^8

A13.

Simplify $\dfrac{10p^3q^5r}{4p^3q^6}$

$$\frac{\cancel{10}^{5}\cancel{p^3}\cancel{q^5}r}{\cancel{4}_{2}\cancel{p^3}q^{\cancel{6}}} \qquad \frac{5r}{2q}$$

$\dfrac{5r}{2q}$

A14.

$$s = \frac{v^2 - u^2}{2a}$$

$v = 7$
$u = 5$
$a = 3$

Work out the value of s.

$$s = \frac{(7)^2 - (5)^2}{2(3)}$$

$$= \frac{49 - 25}{6}$$

$$= \frac{24}{6}$$

$$= 4$$

$s =$ 4

A15.

Make a the subject of the formula $v = u + at$

$$-u \quad -u$$

$$\frac{v - u}{t} = \frac{at}{t}$$

$$a = \frac{v - u}{t}$$

A16.

Make v the subject of the formula $T = \frac{1}{2}mv^2$

$$2T = mv^2$$

$$\frac{2T}{m} = v^2$$

$$\sqrt{\frac{2T}{m}} = v$$

A17.

Make x the subject of the formula $a = \frac{x+4}{x+2}$

$$a(x+2) = x+4$$

$$ax + 2a = x + 4$$

$$-x \qquad -x$$

$$ax - x + 2a = 4$$

$$-2a \qquad -2a$$

$$ax - x = 4 - 2a$$

$$x(a-1) = 4 - 2a$$

$$x = \frac{4-2a}{a-1}$$

A18.

Make x the subject of the formula $\frac{a}{b} = \frac{2x}{x+5}$

$$a(x+5) = 2x(b)$$

$$ax + 5a = 2bx$$

$$5a = 2bx - ax$$

$$5a = x(2b - a)$$

$$x = \frac{5a}{2b - a}$$

A19.

Expand and Simplify $(x+2)(x+4)(x+1)$

$$(x+2)(x^2 + x + 4x + 4)$$

$$(x+2)(x^2 + 5x + 4)$$

$$x^3 + 5x^2 + 4x + 2x^2 + 10x + 8$$

$$x^3 + 7x^2 + 14x + 8$$

A20.

Expand and Simplify $(x+3)(x-1)^2$

$$(x+3)(x-1)(x-1)$$

$$(x+3)(x^2 - x - x + 1)$$

$$(x+3)(x^2 - 2x + 1)$$

$$x^3 - 2x^2 + x + 3x^2 - 6x + 3$$

$$x^3 + x^2 - 5x + 3$$

A21.

Expand and Simplify $(2x-3)(x-2)(3x-1)$

$$(2x-3)(3x^2-x-6x+2)$$

$$(2x-3)(3x^2-7x+2)$$

$$6x^3-14x^2+4x-9x^2+21x-6$$

$$\underline{6x^3-23x^2+25x-6}$$

A22.

Show that $(5x-4)(3x+1)(2x-7) = 30x^3 - 119x^2 + 41x + 28$

for all values of x.

$$(5x-4)(6x^2-21x+2x-7)$$

$$(5x-4)(6x^2-19x-7)$$

$$30x^3-95x^2-35x-24x^2+76x+28$$

$$\underline{30x^3-119x^2+41x+28}$$

Summary Test Biology

This covers sections of 4.1 - Cell Biology (All of 4.1.1 - Cell Structure) from the AQA specification.

Topics tested:

4.1.1.1 Eukaryotes and prokaryotes
4.1.1.2 Animal and plant cells
4.1.1.3 Cell specialisation
4.1.1.4 Cell differentiation
4.1.1.5 Microscopy

Time: 50 minutes
Total Marks: 40

Section A: Core Knowledge (10 marks)

1. State two differences between eukaryotic and prokaryotic cells. (2)

_____

_____

2. Describe the function of: (3)

a) Ribosomes

b) Mitochondria

c) Cell membrane

3. Explain why bacterial cells are described as prokaryotic. (2)

_____

4. Give one example of a specialised animal cell and describe how it is adapted to its function. (3)

_____

Section B: Application and Microscopy (15 marks)

5. A plant cell is observed under a microscope. It contains chloroplasts and a large vacuole.

Explain how these structures help the plant cell function effectively. (4)

_____

6. Calculate the actual size of a cell if the image size is 2.4 mm and magnification is ×600.

Show full working. (3)

7. Explain why electron microscopes provide greater detail than light microscopes. (4)

_____

8. Describe how cell differentiation differs in animals and plants. (4)

_____

Section C: Higher Challenge (15 marks)

9. Compare a bacterial cell and a plant cell in terms of structure and genetic material. (6)

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10. A student observes cells dividing rapidly in the root tip of a plant.

Explain why cells in this region show high levels of differentiation later in development. (4)

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11. A microscope has a ×10 eyepiece and a ×40 objective lens.

a) Calculate total magnification. (1)

b) Explain how increasing magnification can affect resolution. (4)

Mark Scheme

1. Any two: nucleus present in eukaryotes; DNA free in cytoplasm in prokaryotes; prokaryotes smaller; plasmids present. (2)

2. Ribosomes – protein synthesis (1); Mitochondria – aerobic respiration/energy release (1);

Cell membrane – controls movement of substances in and out (1)

3. Genetic material not enclosed in a nucleus. (2)

4. Example e.g. sperm cell – tail for movement; nerve cell – long axon; red blood cell – no nucleus, biconcave. (3)

5. Chloroplasts – photosynthesis (2); Vacuole – maintains turgor/support (2)

6. Convert mm to µm if needed; 2.4 mm = 2400 µm; 2400 ÷ 600 = 4 µm. (3)

7. Electrons have shorter wavelength; higher resolution; greater magnification; more detail visible. (4)

8. Animals differentiate early and are limited; plants differentiate throughout life at meristems. (4)

9. 6-mark comparison: nucleus vs none; plasmids; cell wall composition; chloroplasts absent in bacteria; size differences; cytoplasm/ribosomes. (6)

10. Meristem cells divide then specialise; gene expression changes; adaptation for specific functions. (4)

11. a) 400 (1); b) Magnification does not equal resolution; higher magnification without resolution gives blurred image; resolution = ability to distinguish detail. (4)

Revision in flashcards

Topic 1 - Cell Biology
4.1 Cell biology
4.1.1 Cell structure

4.1.1.1 Eukaryotes and prokaryotes

Q1. Define eukaryotic cells and 2 examples of them.
A1. Larger, complex cells with the **DNA enclosed in the nucleus** examples include **plant and animal cells**.

Q2. Define prokaryotic cells and 1 example of them.
A2. Smaller, simple cells that **don't have a nucleus to keep their DNA** (genetic material) but instead it is stored in the cytoplasm. An example is **bacterial cells**.

Q3. What are the key **organelles** (things inside a cell) of a eukaryotic cell?
A3. Cell membrane, cytoplasm and a nucleus containing DNA.

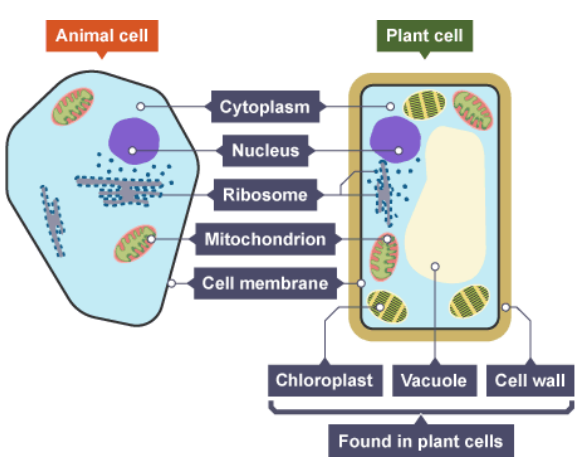
Q4. What are the key organelles (things inside a cell) of a prokaryotic cell?
A4. Cytoplasm, cell membrane, cell wall (in bacterial cells), plasmids (sometimes) and DNA **NOT** in a nucleus but **one single circular loop**.

Q5. What are plasmids?
A5. Although the main DNA is stored in one singular loop in the cytoplasm for a bacterial cell (type of prokaryotic cell). There can be **other smaller loops of 'extra' DNA** called plasmids.

Q6. What are the rough sizes of plant, animal and bacterial cells?
- Plant cells = 10 - 100 μm
- Animal cells = 10 - 30 μm
- Bacterial cells = 0.5 - 5 μm (around 10 times smaller than plant and animal cells)

4.1.1.2 Animal and plant cells

Q7. Draw the general structure of an animal and plant cell labelling all the organelles.
A7.



Q8. Describe the functions of the organelles in a plant and animal cell.
A8.

1. Nucleus
    a. Contains the DNA of the cell.
    b. Acts as the control center (Brain) regulating the growth, division etc of the cell.
2. Cytoplasm
    a. Liquid substance in which chemical reactions occur.
    b. Contains enzymes that speed up the rate of reactions.
    c. Is where other organelles stay.
3. Cell membrane
    a. Controls what enters and leaves the cell.
4. Mitochondria
    a. Where aerobic respiration reactions take place providing energy for the cell (powerhouse of the cell).
5. Ribosomes
    a. Where protein synthesis occurs i.e make proteins for the cell.
    b. They are found on a structure called the rough endoplasmic reticulum.
6. Chloroplasts (plant cell only)
    a. Where photosynthesis takes place.
    b. Contains chlorophyll pigment (makes plant green) harvesting light for photosynthesis.
7. Vacuole (plant cell only)
    a. Contains cell sap.
    b. Improves the cell's rigidness.
8. Cell wall (present in plant and **Algal cells**)
    a. Made of cellulose (a carbohydrate)
    b. Strengthens the cell (helps maintain the cells shape)

Q9. Name the organelles in a bacterial cell (similar to Q4 since bacterial cells are a type of prokaryotic cell).
A9. Cytoplasm, cell membrane, cell wall, single circular strand of DNA, plasmids.

4.1.1.3 Cell specialisation

Q10. Define cell specialisation.
A10. When a cell has specific adaptations (shape, structure and functions) to be able to perform specialized roles in an organism.

Q11. Name 3 specialized animal cells.
A11. We learnt the general structure of animal cells but some specialized animal cells are - Sperm cells, nerve cells and muscle cells.

Q12. Name 3 specialized plant cells.
A12. Root hair cells, xylem cells and phloem cells.

Q13. Explain the function of a sperm, nerve and muscle cell.
A13.

- Sperm cell - Carry the male DNA to the egg cell for reproduction.
- Nerve cell - Transmit electrical signals quickly from one place in the body to the another.
- Muscle cell - Contract quickly to cause movement of the bones and muscles e.g contracting the heart to push blood around the body.

Q14. Explain the function of a root hair, xylem and phloem cell.
A14.

- Root hair cell - Found in the tips of the root of a plant, they take up water and mineral ions from the soil.
- Xylem cell - Transports water and mineral ions from the roots to the shoots of the plant and provides support to the plant.
- Phloem cell - Carries the products of photosynthesis (food / dissolved sugars) to all parts of the plant.

Q15. Explain the specializations / adaptations of a sperm, nerve and muscle cells that help them do their function.
A15.
Sperm cell -

- Long tail (flagellum) → helps it swim to the egg.
- Many mitochondria → provide energy for movement (since it's the POWER house of the cell).
- The top of the head has a cap-like organelle called acrosome → which contains digestive enzymes to penetrate the membrane of the egg cell.

Nerve cell -

- Long axon → allows impulses to travel long distances.
- Branched dendrites (extensions from the cell body) → allow branched connections to be formed with other nerve cells.

Muscle cell -

- Contains protein fibres that slide over each other causing the muscle to contract.
- Many mitochondria provide energy for contraction.

Q16. Explain the specializations / adaptations of a root hair, xylem and phloem cells that help them do their function.
A16.
Root hair cell -

- Large surface area (long hair) → increases absorption.
- Thin cell wall → easier for water to enter.
- Large vacuole - contains cell sap which helps draw water into the root hair cell by osmosis.

Xylem cell -

- Hollow tubes → water flows easily.
- Walls are strengthened with a chemical called lignin → strong support to withstand pressure of water movement.

Phloem cell -

- Tube-like structures (sieve tubes) → allow dissolved sugars to flow through the plant.
- Sieve plates with small holes (formed when the cell wall of each cell breaks down) → allow substances to move between cells.

4.1.1.4 Cell differentiation

Q17. Define cell differentiation.
A17. The process by which a cell changes acquiring different sub-cellular structures to become a specialized cell.

Q18. When does differentiation take place in animal cells **usually**?
A18. Early in development. Once most animal cells specialize they can't differentiate again.

Q19. In **mature** animals, what is the most common function of cell division?
A19. Repair and replacement of damaged cells. This is since mature animals don't usually undergo much growth they don't require new specialized cells.

Q20. True or False - many types of plant cells retain their ability to differentiate throughout life unlike animal cells.
A20. True, many types of plant cells retain their ability to differentiate throughout life.

4.1.1.5 Microscopy

Q21. What is microscopy?
A21. The use of microscopes to study small things like cells that can not be seen with the naked eye.

Q22. How have microscopy techniques developed over time?
A22. Light microscopes to electron microscopes. Initially light microscopes were used to observe tissue and larger sub-cellular structures / organelles. Then came electron microscopes providing greater magnification allowing observation of smaller sub-cellular structures like plasmids, mitochondria, etc.

Q23. What is the difference between magnification and resolution?
A23.

- Magnification is how much bigger the image is compared to the real object.
- Resolution is how clearly two points can be seen as separate. A higher resolution leads to a more detailed and clearer image.
- It's basically that one is about size and the other is about clarity.

Q24. Name the main parts of a light microscope.
A24.

- Eyepiece lens – the lens you look through.
- Objective lenses – lenses close to the specimen that magnify the image.
- Stage – holds the microscope slide.
- Light source / mirror – shines light through the specimen.
- Slide - piece of glass which holds the specimen being observed.
- Coverslip - thin glass placed on top to keep the specimen flat and protect the lens.
- Stain - coloured chemical used to make cell structures easier to see.

Q25. Briefly explain how a light microscope works.
A25.

- Place the specimen on a microscope slide and add a drop of stain (e.g iodine), then place a cover slip.
- Light shines through the specimen from below.
- The objective lens magnifies the image
- The eyepiece lens further magnifies the image for the viewer to observe.

Q26. How do electron microscopes produce an image?
A26. Electron microscopes use beams of electrons instead of light to produce images.

Q27. Why are electron microscopes better than light?
A27. Electrons have a much shorter wavelength than light, so electron microscopes have:

- Much higher magnification
- Much higher resolution

Q28. What is the formula for magnification?
A28.

$$magnification = \frac{size\ of\ image}{size\ of\ real\ object}$$